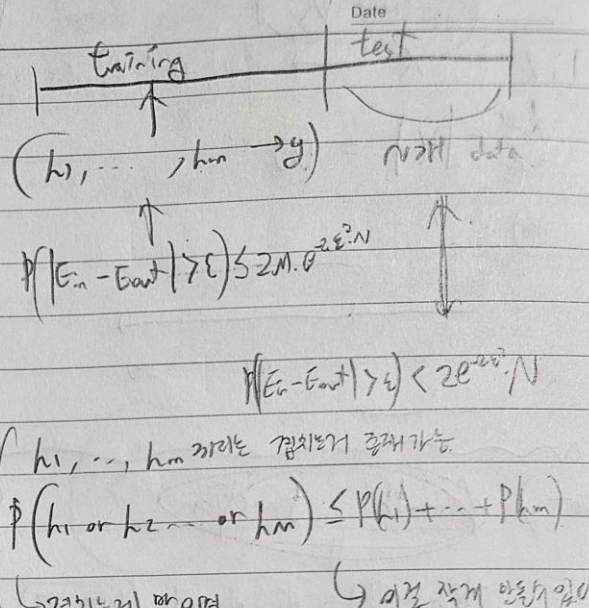
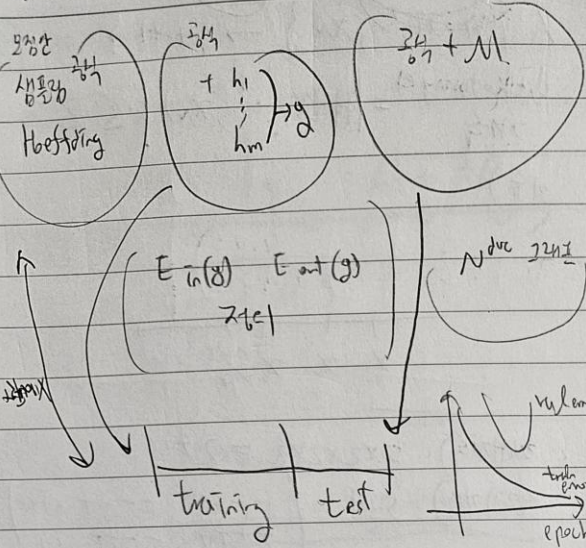


VC dimension



$E_{out}(g)$: g 가 f 의 함수를 예측할 때의 오차

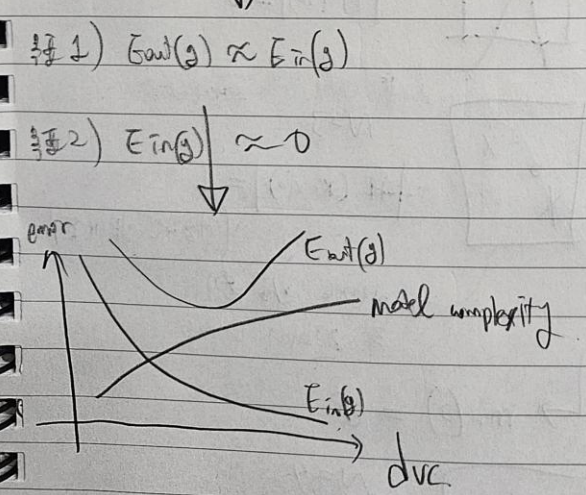
$E_{in}(g)$: g 가 f 의 함수를 예측할 때의 오차

모든 g 에 대해 $E_{out}(g) \approx 0$

VC dimension

$m \rightarrow \infty$ 가능

$n \rightarrow \infty$ 가능



$h(x) = \text{sign} \left(\sum_{i=1}^d w_i x_i - \text{threshold} \right)$

$h(x) = \text{sign} \left(\sum_{i=1}^d w_i x_i + w_0 \right)$

$h(x) = \text{sign} \left(\sum_{i=1}^d w_i x_i \right)$

$h(x) = \text{sign} (w \cdot x)$

VC dimension

$$P(|E_{in}(g) - E_{out}(g)| > \epsilon) \leq 2M e^{-2\epsilon^2 N}$$

N 증가

$h(x) \approx \text{dichotomy}$

but, $\begin{pmatrix} - & + \\ + & - \end{pmatrix}$ 이진

이진

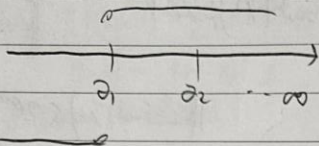
ibis

growth

$$x \rightarrow \{-1, +1\}$$

$$|H| = \infty$$

이유)



Def) Growth function

가장 최대 dichotomy (N개 기준)

$$m_H(N) = \max_{x_1, \dots, x_N \in \mathcal{X}} |f(x_1, \dots, x_N)|$$

$$m_H(N) \leq 2^N$$

$$m_H(N) \leq 2^k$$

$$m_H(N) \leq \sum_{i=0}^{k-1} \binom{N}{i} \approx N^k \approx N^{k+1}$$

$\rightarrow m_H(N)$ 이 bound 된다는 의미

$$f) N(2) = \frac{N(N-1)}{2} \approx N^2$$

$$m_H(N) = N+1 \approx N$$

$$m_H(N) = \frac{N^2}{2} + \frac{N}{2} + 1 \approx N^2$$

$$m_H(N) = 2^N \approx 2^N$$

$$N < N^2 \ll 2^N$$

$N \rightarrow m_H(N)$ 은 차등화

Vs

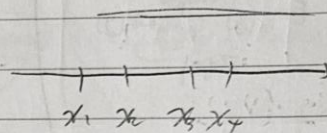
dichotomy

(정규화)

$$x_1, x_2, \dots, x_N \rightarrow \{-1, +1\}$$

$$dichotomy = |H(x_1, \dots, x_N)| \leq 2^N$$

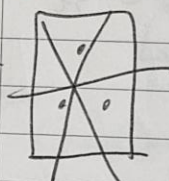
이유)



$$2 \times 2 \times 2 \times 2 = 2^4$$

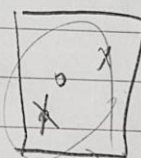
$$5 \text{개} \begin{pmatrix} - & - & - & - \\ + & + & + & + \end{pmatrix}$$

$$m_H(N) \text{ 계산}$$



$$N=3$$

$$|H(\dots)| = 8$$



$$N=3$$

$$|H(x, x, x)| = 6$$

(계산하면 이렇게 됨)

이런게는 N 분기

즉 2^3에서 배제됨

$$m_H(3) = 8$$



$$N=4$$

$$|H(x, x, x, x)| = 14$$

(계산하면 이렇게 나옴)

이런게는 N 분기

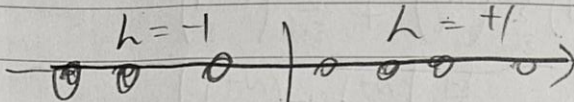
즉 2^4에서 배제됨

$$m_H(4) = 14 \neq 2^4$$

break point of $\mathcal{H} = k$

④ positive rays

$$m_H(N) = N+1$$

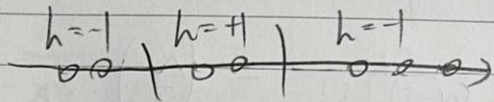


$$h: \mathbb{R} \mapsto \{-1, +1\}$$

$$h(x) = \text{sign}(x-a)$$

⑤ positive intervals

$$m_H(N) = \left(\frac{N+1}{2}\right) + 1 = \frac{N^2}{2} + \frac{N}{2} + 1$$



$$h: \mathbb{R} \mapsto \{-1, +1\}$$

⑥ Convex set

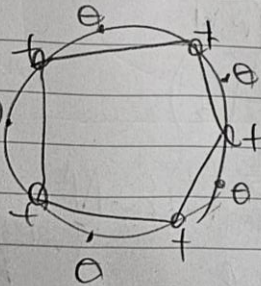
$$m_H(N) = 2^N$$

$$h: \text{circle} \mapsto \{-1, +1\}$$

$$x^2 + y^2 = 1$$

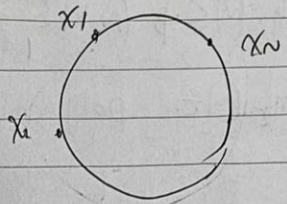
$$h(x) = +1 \quad (\text{if } x)$$

shattered



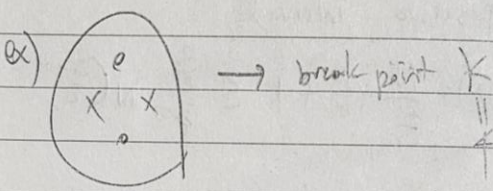
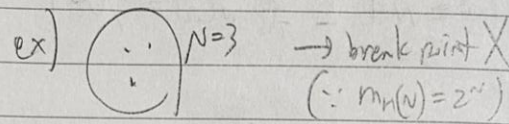
$$x_1, \dots, x_N$$

$$\begin{bmatrix} \oplus \\ \ominus \end{bmatrix}$$



$$m_H(N) < 2^k$$

→ shattered 안됨



$$m_H(4) < 2^4$$

$$\begin{matrix} 2D \text{ Perceptron} & k=4 \\ 3D \text{ perceptron} & k=5 \\ M \cdot D \text{ Perceptron} & k=M+2 \end{matrix}$$

$$m_H(N) = N+1$$

$$m_H(2) = 3 < 2^2$$

Positive rays → k=2

$$m_H(N) = \frac{N^2}{2} + \frac{N}{2} + 1$$

$$m_H(3) = 7 < 2^3$$

positive intervals → k=3

$$m_H(N) = 2^N$$

$$m_H(k) = 2^k$$

→ break point X
k=∞

$$P(|\bar{E}_n - E_n| > \epsilon) \leq 2M \cdot e^{-2\epsilon^2 N}$$

$m_H(N)$: polynomial 양쪽 3. 명제

$$③ m_H(N) \leq \sum_{i=0}^{k-1} \binom{N}{i} \sim N^{k-1}$$

→ $m_H(N)$ is polynomial

ex) positive ray

$$m_H(N) = N+1 \leq \sum_{i=0}^1 N \zeta_i = N+1$$

ex) positive interval

$$m_H(N) = \frac{N^2}{2} + \frac{N}{2} + 1 \leq \sum_{i=0}^2 N \zeta_i$$

ex) 2D perception

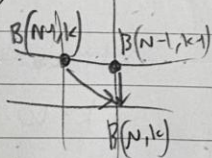
$$m_H(N) \leq \sum_{i=0}^3 N \zeta_i$$

ps) $m_H(N) \leq \dots \leq \text{polynomial}$
 $\parallel$
 $B(N, k)$

$$m_H(N) \leq B(N, k) \leq \text{polynomial}$$

growth function
 $\parallel$
 difference of
 dimension
 $(N, k+1)$

lemma) $B(N, k) \leq B(N-1, k) + B(N-1, k-1)$



		k					
		1	2	3	4	5	6
$m_H(k) \zeta^0$	1	1	2	2	2	2	2
$m_H(k) \zeta^1$	2	1	3	4	4	4	4
$m_H(k) \zeta^2$	3	1	4	7	8	8	8
$m_H(k) \zeta^3$	4	1	5	11	11	11	11
N	5	1	6	1	1	1	1
	6	1	7	1	1	1	1

$$H(N, k) = \sum_{i=0}^{k-1} \binom{N}{i} \text{ 일 때,}$$

$$H(N, k) = H(N-1, k) + H(N-1, k-1)$$

$$\text{or) } \sum_{i=0}^{k-1} \binom{N-1}{i} + \sum_{i=0}^{k-2} \binom{N-1}{i}$$

$$= 1 + \sum_{i=1}^{k-1} \left[\binom{N-1}{i} + \binom{N-1}{i-1} \right]$$

$$\Rightarrow \frac{(N-1)!}{(N-1-i)! i!} + \frac{(N-1)!}{(N-1-i)! (i-1)!}$$

$$\Rightarrow \frac{(N-1)! (N-i+1)}{(N-1-i)! i!}$$

$$\Rightarrow \frac{N!}{(N-i)! i!} = \binom{N}{i}$$

$$= 1 + \sum_{i=1}^{k-1} \binom{N}{i}$$

$$= \sum_{i=0}^{k-1} \binom{N}{i} = H(N, k)$$

→ $B(N, k) \leq H(N, k)$

$$B(N, k) \leq B(N-1, k) + B(N-1, k-1)$$

$$\hookrightarrow H(N-1, k) + H(N-1, k-1)$$

$$\hookrightarrow H(N, k)$$

$$\hookrightarrow \sum_{i=0}^{k-1} \binom{N}{i}$$

$$m_H(N) \leq B(N, k) \leq \sum_{i=0}^{k-1} \binom{N}{i} \sim N^{k-1}$$

$$m_H(N) \leq \text{polynomial} \quad \square$$

0123

$$P(|E_{in}(g) - E_{out}(g)| > \epsilon) \leq 2 \sum_{i=0}^M \binom{N}{i} e^{-\epsilon^2 i^2 / N}$$

$$\leq \sum_{i=0}^k \binom{N}{i} \approx N^k / N^{dvc}$$

$$N \leq dvc(H) \leq k$$

$$P(|E_{in}(g) - E_{out}(g)| > \epsilon) \leq 4 m_H(2N) \cdot e^{-\frac{\epsilon^2}{8} N}$$

Vapnik-Chervonenkis inequality.

Def) V_C dimension

$dvc(H) = \text{shatter}$ 할 수 있는 가장 큰 수.

$$N \leq dvc(H) \leq k$$
 k가 H 의 break point 일때
 H 가 N 의 shatter할 수 있는 일대

$$P(|E_{in}(g) - E_{out}(g)| > \epsilon) \leq m_H(2N) e^{-\frac{\epsilon^2}{8} N}$$

$$N^{dvc} \cdot e^{-N} \approx 4 \cdot N^{dvc} \cdot e^{-\frac{\epsilon^2}{8} N}$$

$$\approx \frac{N^d}{e^N}$$

$$\left(N^d \ll e^N \right)$$
 (가장 중요)

ex) M-D perceptron,
 $dvc(H) < k = M+2$
 $dvc(H) \leq M+1$

$$m_H(N) \leq \sum_{i=0}^k \binom{N}{i} \approx N^k$$

$$m_H(N) \leq \sum_{i=0}^{dvc} \binom{N}{i} \approx N^{dvc}$$

$$\delta = 4 m_H(2N) \cdot e^{-\frac{\epsilon^2}{8} N}$$

$$\rightarrow \epsilon = \sqrt{\frac{8}{N} \ln \frac{4 m_H(2N)}{\delta}}$$

$$P(|E_{out} - E_{in}| < \epsilon) \geq 1 - \delta$$

$$m_H(N) \leq \sum_{i=0}^{dvc} \binom{N}{i} \approx N^{dvc}$$

$$\leq \frac{1}{(1-\frac{dvc}{N})} + \frac{N}{(1-\frac{dvc}{N})} + \sum_{i=2}^{dvc} N^{dvc}$$

$$\leq (dvc) \cdot N^{dvc}$$

ex) Positive ray
 $dvc=1 < k=2$

$$E_{out} - E_{in} \leq \epsilon = \sqrt{\frac{8}{N} (dvc \ln(2N) + \ln(1/\delta))}$$

ex) 2-D perceptron,
 $dvc=3 < k=2+2=4$

$$P(E_{out} \leq E_{in} + \sqrt{\frac{8}{N} (2 dvc \ln(2N) + \ln \delta)})$$

ex) Convex set
 $dvc = \infty$

$\geq 1 - \delta$
 SS
 90%

Bolton's inequality

Date

Learning task

input: X

output: $y = t = \text{target}$

target function $f: X \rightarrow Y$

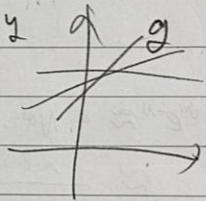
이항적인 AI 문제

Data: $(x_1, y_1), \dots, (x_n, y_n)$

data set X
input

hypothesis $g: X \rightarrow Y$

||
만들어진 AI 문제
||
이항적



이항적인 f

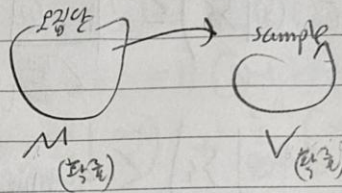
training data

학습
데이터

$g \approx f$

hypothesis set
 $\mathcal{H} = \{h_1, h_2, \dots, h_m\}$

$\mathcal{H} = \{h_1, h_2, \dots, h_m\}$
 $g \in \mathcal{H}$



$V \subseteq M$ 를 설명가능하?

(a) 'possible' 관계 N_o

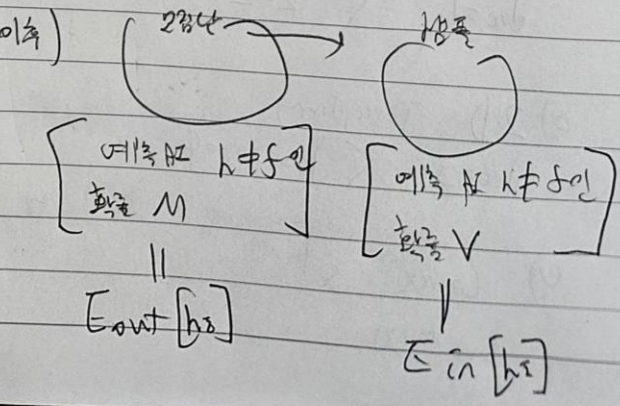
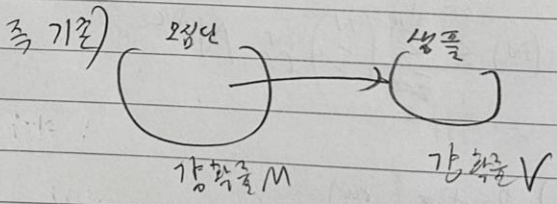
(c) 'probable' 관계 Yes

M : 이가 틀리는 비율

$f: X \rightarrow Y$ (learning task)
||
이항적

이항적인
정답 AI 문제

$L(x) = \text{만들어진 AI 문제} = \text{fixed}$



ex) Coin analogy

M : M 개의 동전을 1.번 던질 때의
그 중 한 번이라도
10번이나 HH..H 일 확률.

Q.) $n=1$ (sample=1)

$\rightarrow p \approx 0.1 \rightarrow$ 낮게 나옴

Q) $n=1000$ (sample=1000)

$\rightarrow p \approx 0.63 \rightarrow$ 2가지.

즉, 같은 AI 모델인데도

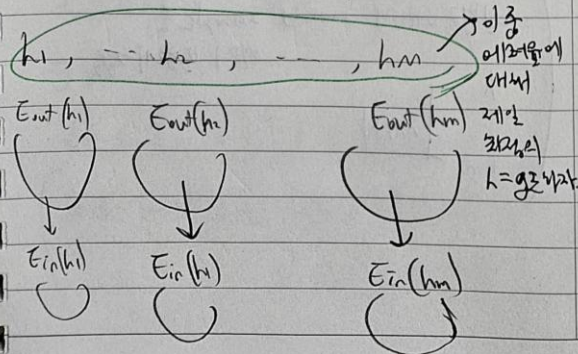
Sample에 따라 (성능)=에러율이
다르게 나옴.

but 실제로는 같은 M 니까

(모집단에서) 에러율이 같음.

$$P(|U - \mu| > \epsilon) \leq 2e^{-2\epsilon^2 \cdot N}$$

공식 안다.



PAC Learning.

Date

$$P(E_{in}(\theta) - E_{out}(\theta) > \epsilon) \\ \leq P(E_{in}(h) - E_{out}(h) > \epsilon \\ \text{or } |E_{in}(h_1) - E_{out}(h_1)| > \epsilon \\ \text{or } |E_{in}(h_m) - E_{out}(h_m)| > \epsilon)$$

$$\leq \sum_{m=1}^M P(E_{in}(h_m) - E_{out}(h_m) > \epsilon)$$

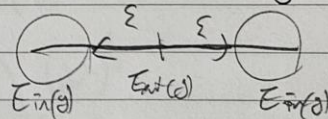
(각각마다 다르게 있다)

$$\leq \sum_{m=1}^M P(|E_{in}(\theta) - E_{out}(\theta)| > \epsilon)$$

$$\leq \sum_{m=1}^M P(E_{in}(h_m) - E_{out}(h_m) > \epsilon)$$

$$\leq \sum_{m=1}^M 2e^{-2\epsilon^2 \cdot N}$$

$$\therefore P(|E_{in}(\theta) - E_{out}(\theta)| > \epsilon) \leq 2 \cdot M \cdot e^{-2\epsilon^2 \cdot N}$$



$$\begin{cases} \text{Sample} \geq 2\epsilon^2 N & P(\text{sample} - 2\epsilon^2 N > \epsilon) \\ \text{Sample} < 2\epsilon^2 N & P(2\epsilon^2 N - \text{sample} > \epsilon) \end{cases}$$

$$\rightarrow P(E_{out}(\theta) - E_{in}(\theta) > \epsilon) < \delta \approx 1\%$$

$$\log \delta = -2\epsilon^2 \cdot N + \log(2M)$$

$$\rightarrow \epsilon = \sqrt{\frac{1}{2N} \cdot (-\log \delta + \log(2M))}$$

$$P(E_{out}(\theta) \leq E_{in}(\theta) \leq \epsilon) \geq 1 - \delta \approx 99\%$$

$$\rightarrow P(E_{out}(\theta) \leq E_{in}(\theta) \pm \epsilon) \geq 1 - \delta \approx 99\%$$

RAC Learning 3월.

Date

$$P(E_{out}(g) \leq E_{in}(g) + \epsilon) \geq 1 - \delta$$

$$P\left(\underbrace{E_{out}(g)}_{\text{generalization error}} \leq \underbrace{E_{in}(g)}_{\text{sample error}} + \sqrt{\frac{1}{2N} (\log \delta + \log(m))}\right) \geq 1 - \delta \approx 90\%$$

• ϵ 에는 N 에 반비례

$$\therefore N > (-\log \delta) > \log M$$

• Model 복잡도

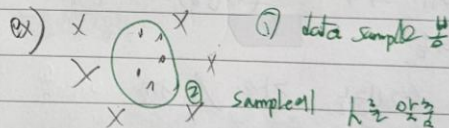
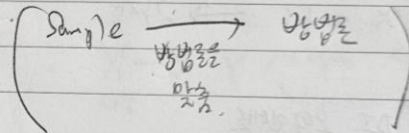
$$N \uparrow \rightarrow \epsilon \downarrow$$

$$M \uparrow \rightarrow \epsilon \uparrow \quad M \downarrow \rightarrow \epsilon \downarrow$$

$$\delta \downarrow \rightarrow \epsilon \uparrow$$

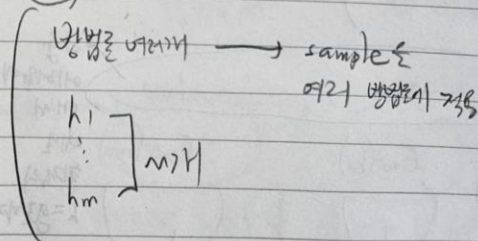
Data Snooping

부정확한
실무적 문제



Hoetting Inequality 적용 불가.

기준



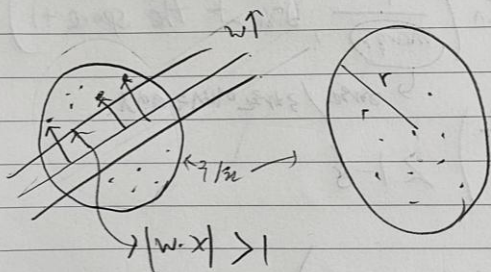
VC dimension of Hyperplanes

$$\text{let } S = \{x \mid \|x\| \leq 1\}$$

Then, VC-dimension d of the set of canonical hyperplanes

$$x \mapsto \text{sgn}(w \cdot x) \mid \min_{x \in S} |w \cdot x| = 1 \mid \|w\| \leq \Lambda$$

$$d \leq r^2 \Lambda$$



$$\frac{1}{\|w\|}, \min_{x \in S} |w \cdot x| = 1 \implies \|w\| \leq \Lambda$$

가장 작은 값

$\|w\|$ 가 큰 값의 최댓값

$\rightarrow \Lambda$ 가 최댓값

P.S) Assume $\{x_1, \dots, x_d\}$ is fully shattered

$$\exists y = (y_1, \dots, y_d) \in \{-1, +1\}^d$$

$\exists w$ s.t.

$$\forall i \in [1, d], 1 \leq y_i(w \cdot x_i)$$

$$\left(\begin{aligned} d &\leq w \cdot \sum_{i=1}^d y_i x_i \leq \|w\| \cdot \left\| \sum_{i=1}^d y_i x_i \right\| \\ &\leq \Lambda \left\| \sum_{i=1}^d y_i x_i \right\| \end{aligned} \right)$$

$\forall y \in \{-1, +1\}^d$ hold.

It also hold on

$\{y_1, \dots, y_d\}$ drawn iid

according to a uniform distribution over $\{-1, +1\}$

For $i \neq j$, we have

$$E[y_i y_j] = E[y_i] E[y_j] = 0$$

$$E[y_i^2] = 1$$

$$\begin{aligned} d &\leq \Lambda \cdot E \left[\left\| \sum_{i=1}^d y_i x_i \right\|^2 \right] \\ &\leq \Lambda \cdot E \left[\sum_{i=1}^d \|x_i\|^2 \right] \\ &= \Lambda \cdot \sum_{i=1}^d E[y_i^2] (x_i \cdot x_i) \\ &= \Lambda \cdot \sum_{i=1}^d (x_i \cdot x_i) \end{aligned}$$

$$\leq \Lambda \cdot (dr)^2$$

$$\Lambda \cdot r \cdot \sqrt{d}$$

$$\therefore \sqrt{d} \leq \Lambda \cdot r$$

$$\exists \frac{1}{2} P(E_{out} \leq E_{in} + \sqrt{\frac{8}{N} \left(\sum_{i=1}^N \log \frac{1}{\delta} \right)}) \geq 1 - \delta$$

$$r^2 \Lambda^2$$

ibis

VC of Break Point

2D perception, $k=4$

3D perception, $k=5$

M-D perception, $k=M+2$

$$\text{VC}(\mathcal{H}) \leq M+1$$

$$\gamma \leq \min \left(\frac{r^2}{\text{Margin}}, \dim \text{ of the space} + 1 \right) = \min \left(\frac{r^2}{\text{Margin}}, \dim \text{ of the space} + 1 \right)$$

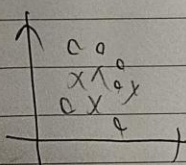
kernel 넣을수록
마진 점점 커짐.

↪ 2차원 / 3차원 이어서 4차원.

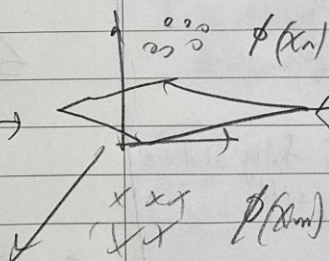
$$P \left(E_{\text{out}} \leq E_{\text{in}} + \sqrt{\frac{2}{N} (\text{VC}(\mathcal{H}) - \ln \delta)} \right) \geq 1 - \delta$$

$$\min \left(\frac{r^2}{\text{margin}}, \dim \text{ of the space} + 1 \right)$$

2) SVM



$X \rightarrow \phi(X)$



$$w_0 \phi_0(x) + \dots + w_n \phi_n(x) = 0$$